Reproducibility

Reproducibility I

Reproducibility in computational research has well-documented prerequisites: open data, open code, and a deterministic build that can be re-run from scratch [@peng2011reproducible]. The bundled `manuscript/config.yaml` is intentionally configured to satisfy all three for **strict reproducibility**:

1. `project_config.search.sources: [local]` consumes da ta/corpus.json, which is a curated and committed JSON corpus. No network is required to run the pipeline.
2. `search.cache_dir: output/search/cache` writes deterministic JSON cache files; running the same query twice produces a byte-identical artifact tree (modulo timestamp metadata in the cache file itself).
3. `enrichment.fetch_abstracts: true` reads abstracts directly from the corpus when present; no network fetch is required.

Reproducibility II

4. `enrichment.fetch_fulltext: false` is the default — full-text fetching is opt-in and gated behind the optional `pypdf` dependency (`uv sync --group rendering`).
5. `llm.enabled: false` is the default — the LLM stage is opt-in and requires a running `ollama serve`. When enabled, `seed: 42` and `temperature: 0.0` are pinned.

Switching to live search I

Replace the `sources` list with the desired backend set:

```yaml
search:
  query: "your topic"
  sources: [arxiv, crossref]
  crossref_mailto: "you@example.org"
```

A first live run populates `output/search/cache/`. Commit that directory to the repo and the pipeline becomes reproducible across machines without further configuration changes.

Determinism guarantees I

The full determinism contract is itemised in [@tbl:determinism]: every pipeline stage is annotated as fully, conditionally, or non-deterministic, with an explicit mechanism column so reviewers can audit each row independently.

Table 1: Determinism contract by pipeline stage. Cached stages are byte-stable across reruns; live stages depend on the upstream source and are pinned to the cache file once a successful run completes.

Stage	Deterministic?	Mechanism
Search (cached hit)	yes	`SearchCache` JSON files
Search (cache miss)	no	live API
Dedup / merge	yes	DOI / arXiv-id canonical keys; tie-break by score then year
Citation-key generation	yes	unicode folding + stop-word skip; collision suffix is deterministic
BibTeX writer	yes	byte-stable format pinning (verified by `tests/infra_tests/reference/`)
Abstract fetch	yes (cached) / no (live)	per-paper `<safe_id>.txt` cache
Fulltext fetch	yes (cached) / mostly (live)	per-paper `<safe_id>.{pdf,txt}` cache; live fetch's `pypdf` text extraction is not bit-stable across versions
LLM synthesis	mostly	`seed=42`, `temperature=0.0`; Ollama deterministic up to its own minor variance

Determinism guarantees II

Figure generation	yes (within Matplotlib version)	fixed palette, fixed bin width, no random subsampling

Verifying reproducibility locally I

```
# Run twice; nothing in output/ should diff except the cac
uv run python projects/templates/template_search_project/s
mv projects/templates/template_search_project/output proje
uv run python projects/templates/template_search_project/s
diff -ru \
    projects/templates/template_search_project/output_firs
    projects/templates/template_search_project/output/corp
```

The only expected differences are inside `output/search/cache/search_*.json`, where `_cached_at` is wall-clock time at write.

Limitations I

The reproducibility contract enumerated in [@sec:reproducibility] (items 1–5 and [@tbl:determinism]) does not eliminate the following well-defined sources of non-reproducibility, which are surfaced here so reviewers can audit them explicitly rather than inferring from the contract table:

- **Live search drift.** When `project_config.search.sources` includes `arxiv` or `crossref`, the first cache-miss invocation hits the live API; the cached JSON freezes that response, but two cold-start clones running on different days will see different paper sets. Pin a `LocalBackend` corpus or commit o `utput/search/cache/` to break this dependency.

Limitations II

- **`pypdf` version drift.** The fulltext fetcher uses `pypdf` to extract text from a downloaded PDF. `pypdf`'s text-extraction algorithm is not bit-stable across major versions; upgrading `pypdf` can produce different `<safe_id>.txt` cache contents from the same source PDF. The PDF bytes themselves are bit-stable so the cache freezes the inputs, not the extraction.
- **Ollama version drift.** Pinning `seed=42` and `temperature=0.0` controls Ollama's sampling, but the model weights, tokenizer, and template can change between Ollama releases. Document the Ollama version alongside `config.llm.model` when archiving a run for replication.
- **Paperclip backend status.** The `paperclip` backend is opt-in and the production endpoint may reject the adapter's request method; the run records the response in `SearchResult.errors[paperclip]` and continues. Treat `paperclip` results as advisory until the upstream service stabilises.

Limitations III

- **External backend behaviour outside this project's control.** arXiv and Crossref are the source of truth; this project is faithful to whatever they return. A retraction, metadata fix, or DOI assignment upstream will alter the cache on the next cold-start invocation.

These limitations bound *what* the cache + seed + corpus pinning achieves. Inside those bounds, the contract in [@tbl:determinism] is total: every cached pipeline stage is byte-stable across reruns (verified by `tests/test_pipeline.py::TestRunLiteraturePipeline::test_bibtex_byte_identical_across_reruns`).